Network health · Spring 2025

Ridgeline Schools Weather Network | 1 March–11 April 2025 | 8 stations | Coordinator action brief

> **Decision:** Use clean sections of the weather records, not entire stations indiscriminately. Set aside marked intervals and treat sensor upgrades, pressure calibration, and logger time as separate uncertainties. **Send the volunteer to S06 first** to resolve a broad timestamp integrity problem.

8	180	214	200
stations reviewed	extra duplicate logger rows	missing expected time slots	flag episodes; not bad-row count

What can go into the spring summary?

Station	Rows	Duplicates / gaps	Flagged weather	Coordinator instruction
S01 · Saddle Ridge Hut	6,048	0 / 0	—	Weather usable with normal spot QA; GPS void fixes excluded only as metadata
S02 · North Mesa School	6,048	0 / 0	pressure_hpa	Use other channels; withhold pressure after board replacement pending reference comparison
S03 · Valley Library	6,048	0 / 0	temp_c	Exclude flagged temperature intervals; remaining channels provisionally usable
S04 · Foothills Community Garden	6,048	0 / 0	—	Rain on 2 Apr needs tipping-bucket and regional corroboration
S05 · Eastside Rec Centre	6,048	0 / 0	rh_pct	Withhold affected RH; verify GPS installation coordinates before location-sensitive use
S06 · South Hills Fire Station	6,048	36 / 0	pressure_hpa, temp_c	Withhold clock-affected records until timestamps can be recovered against GPS
S07 · Riverbend Farm	6,048	0 / 0	—	Wind calibration after 26 Mar is unverified; compare to nearby reference
S08 · Plains Grange Hall	5,978	144 / 214	—	Use only unique timestamps; omit outage; review replayed day

Technician visit: S06 · 60 minutes

Why: After the 24 March battery swap, the GPS/logger UTC discrepancy persists; duplicate logger times overlap the restart. This compromises ordering and the interpretation of multiple weather channels. **Second choice: S05** has extensive RH saturation and a persistent GPS installation mismatch, but its remaining weather-channel timing is intact.

0–10 min: Photograph UTC clock, GPS lock/status and power readings without resetting.
10–25: Compare raw NMEA UTC with logger UTC and timezone configuration; check firmware clock-source setting.
25–40: Inspect battery connections, reboot log and buffering/replay settings.
40–55: Take synchronized readings against an independent UTC clock and retain raw logs.
55–60: Record the offset, decide whether timestamp correction is independently demonstrable, and document follow-up.

Important incidents

S03: 481 values stuck at 6.5°C (18–21 Mar), plus isolated large spikes. Do not average the flatline into climate summary.
S05: 451 relative-humidity readings above 100%. Check sensor; do not clip to 100%. Valid NMEA positions systematically disagree with the installation sheet.
S08: 214 missing slots around 16–18 Mar and 144 extra repeated timestamps on 5 Apr. Neither can be repaired by naive interpolation.
S02: pressure step after 21 Mar board exchange is consistent with a calibration discontinuity; needs independent standard.
S04: large rain rates near funnel cleaning on 2 Apr; **keep provisionally** pending independent comparison.
S07: replacement anemometer brand may have a different transfer function. Verify calibration before comparisons.

Method / limitations. Automated range, continuity, persistence, abrupt-jump and NMEA consistency checks; one row in flags.csv represents an episode, not a count of corrupted samples. GPS void records are flagged on metadata only, not grounds to discard coincident weather. The station install sheet is a reference, not guaranteed correct. No replacement values are silently inserted. Thresholds are conservative heuristics; these are provisional station/channel decisions rather than full certified validation.